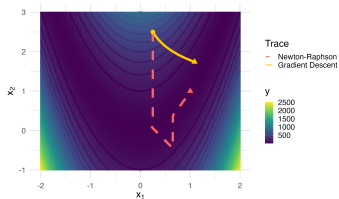
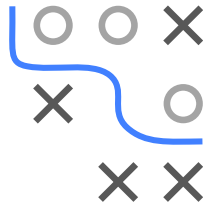


Second order methods

Quasi-Newton



- Newton-Raphson vs. Quasi-Newton
- SR1
- BFGS



QUASI-NEWTON: IDEA

- Start point of **QN method** is (as with NR) a Taylor approximation of the gradient, except that H is replaced by a **pd** matrix $\mathbf{A}^{[t]}$:

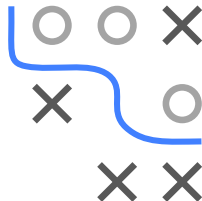
$$\nabla f(\mathbf{x}) \approx \nabla f(\mathbf{x}^{[t]}) + \nabla^2 f(\mathbf{x}^{[t]})(\mathbf{x} - \mathbf{x}^{[t]}) = \mathbf{0} \quad \text{NR}$$

$$\nabla f(\mathbf{x}) \approx \nabla f(\mathbf{x}^{[t]}) + \mathbf{A}^{[t]}(\mathbf{x} - \mathbf{x}^{[t]}) = \mathbf{0} \quad \text{QN}$$

- The update direction:

$$\mathbf{d}^{[t]} = -\nabla^2 f(\mathbf{x}^{[t]})^{-1} \nabla f(\mathbf{x}^{[t]}) \quad \text{NR}$$

$$\mathbf{d}^{[t]} = -(\mathbf{A}^{[t]})^{-1} \nabla f(\mathbf{x}^{[t]}) \quad \text{QN}$$



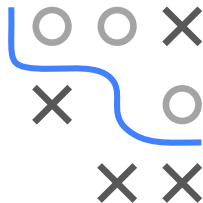
QUASI-NEWTON: IDEA

- Select a starting point $\mathbf{x}^{[0]}$ and initialize pd matrix $\mathbf{A}^{[0]}$ (can also be a diagonal matrix - a very rough approximation of Hessian)
- Calculate update direction by solving

$$\mathbf{A}^{[t]} \mathbf{d}^{[t]} = -\nabla f(\mathbf{x}^{[t]})$$

and set $\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} + \alpha^{[t]} \mathbf{d}^{[t]}$ (Step size through backtracking)

- Calculate an efficient update $\mathbf{A}^{[t+1]}$, based on $\mathbf{x}^{[t]}$, $\mathbf{x}^{[t+1]}$, $\nabla f(\mathbf{x}^{[t]})$, $\nabla f(\mathbf{x}^{[t+1]})$ and $\mathbf{A}^{[t]}$



QUASI-NEWTON: IDEA

- Usually the matrices $\mathbf{A}^{[t]}$ are calculated recursively by performing an additive update

$$\mathbf{A}^{[t+1]} = \mathbf{A}^{[t]} + \mathbf{B}^{[t]}$$

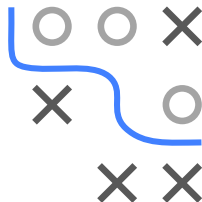
- How $\mathbf{B}^{[t]}$ is constructed is shown on the next slides

- Requirements** for the matrix sequence $\mathbf{A}^{[t]}$:
 - Symmetric pd, so that $\mathbf{d}^{[t]}$ are descent directions
 - Low computational effort when solving LES

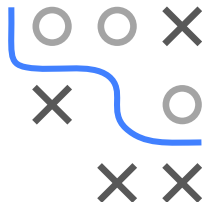
$$\mathbf{A}^{[t]} \mathbf{d}^{[t]} = -\nabla f(\mathbf{x}^{[t]})$$

- Good approximation of Hessian: The “modified” Taylor series for $\nabla f(\mathbf{x})$ (especially for $t \rightarrow \infty$) should provide a good approximation

$$\nabla f(\mathbf{x}) \approx \nabla f(\mathbf{x}^{[t]}) + \mathbf{A}^{[t]}(\mathbf{x} - \mathbf{x}^{[t]})$$



SYMMETRIC RANK 1 UPDATE (SR1)



- Simplest approach: symmetric rank 1 updates (**SR1**) of form

$$\mathbf{A}^{[t+1]} \leftarrow \mathbf{A}^{[t]} + \mathbf{B}^{[t]} = \mathbf{A}^{[t]} + \beta \mathbf{u}^{[t]} (\mathbf{u}^{[t]})^T$$

with appropriate vector $\mathbf{u}^{[t]} \in \mathbb{R}^n$, $\beta \in \mathbb{R}$

SYMMETRIC RANK 1 UPDATE (SR1)

- **Choice of $\mathbf{u}^{[t]}$:** Vectors should be chosen so that the “modified” Taylor series corresponds to the gradient:

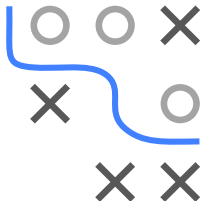
$$\nabla f(\mathbf{x}) \stackrel{!}{=} \nabla f(\mathbf{x}^{[t+1]}) + \mathbf{A}^{[t+1]}(\mathbf{x} - \mathbf{x}^{[t+1]})$$

$$\nabla f(\mathbf{x}) = \nabla f(\mathbf{x}^{[t+1]}) + \left(\mathbf{A}^{[t]} + \beta \mathbf{u}^{[t]}(\mathbf{u}^{[t]})^T \right) \underbrace{(\mathbf{x} - \mathbf{x}^{[t+1]})}_{:= \mathbf{s}^{[t+1]}}$$

$$\underbrace{\nabla f(\mathbf{x}) - \nabla f(\mathbf{x}^{[t+1]})}_{\mathbf{y}^{[t+1]}} = \left(\mathbf{A}^{[t]} + \beta \mathbf{u}^{[t]}(\mathbf{u}^{[t]})^T \right) \mathbf{s}^{[t+1]}$$

$$\mathbf{y}^{[t+1]} - \mathbf{A}^{[t]} \mathbf{s}^{[t+1]} = \left(\beta (\mathbf{u}^{[t]})^T \mathbf{s}^{[t+1]} \right) \mathbf{u}^{[t]}$$

- For $\mathbf{u}^{[t]} = \mathbf{y}^{[t+1]} - \mathbf{A}^{[t]} \mathbf{s}^{[t+1]}$ and $\beta = \frac{1}{(\mathbf{y}^{[t+1]} - \mathbf{A}^{[t]} \mathbf{s}^{[t+1]})^T \mathbf{s}^{[t+1]}}$ the equation is satisfied



SYMMETRIC RANK 1 UPDATE (SR1)

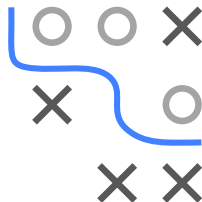
Advantage

- Provides a sequence of **symmetric pd** matrices
- Matrices can be inverted efficiently and stable using Sherman-Morrison:

$$(\mathbf{A} + \beta \mathbf{u}\mathbf{u}^T)^{-1} = \mathbf{A} + \beta \frac{\mathbf{u}\mathbf{u}^T}{1 + \beta \mathbf{u}^T \mathbf{u}}$$

Disadvantage

- The constructed matrices are not necessarily pd, and the update directions $\mathbf{d}^{[i]}$ are therefore not necessarily descent directions



BFGS ALGORITHM

- Instead of Rank 1 updates, the **BFGS** procedure (published simultaneously in 1970 by Broyden, Fletcher, Goldfarb and Shanno) uses rank 2 modifications of the form

$$\mathbf{A}^{[t]} + \beta_1 \mathbf{u}^{[t]}(\mathbf{u}^{[t]})^T + \beta_2 \mathbf{v}^{[t]}(\mathbf{v}^{[t]})^T$$

with $\mathbf{s}^{[t]} := \mathbf{x}^{[t+1]} - \mathbf{x}^{[t]}$

- $\mathbf{u}^{[t]} = \nabla f(\mathbf{x}^{[t+1]}) - \nabla f(\mathbf{x}^{[t]})$
- $\mathbf{v}^{[t]} = \mathbf{A}^{[t]} \mathbf{s}^{[t]}$
- $\beta_1 = \frac{1}{(\mathbf{u}^{[t]})^T (\mathbf{s}^{[t]})}$
- $\beta_2 = -\frac{1}{(\mathbf{s}^{[t]})^T \mathbf{A}^{[t]} \mathbf{s}^{[t]}}$
- The resulting matrices $\mathbf{A}^{[t]}$ are positive definite and the corresponding quasi-Newton update directions $\mathbf{d}^{[t]}$ are actual descent directions

